1. Introduction

1.1 Purpose of the Document

This Software Design Specification (SDS) provides a comprehensive and technically structured description of the Smart Restaurant Point of Sale (POS) System. It outlines the system architecture, design rationale, modules, internal data structures, workflows, and interaction patterns. The document ensures that developers, testers, and stakeholders have a unified understanding of the system's functional and structural blueprint.

1.2 Scope

The Smart Restaurant POS System is designed to streamline restaurant operations by integrating order processing, menu management, billing, table management, inventory control, and user administration. The system supports both on-premises and online operations, offering an intuitive interface for staff and administrators while maintaining data integrity, accuracy, and efficiency.

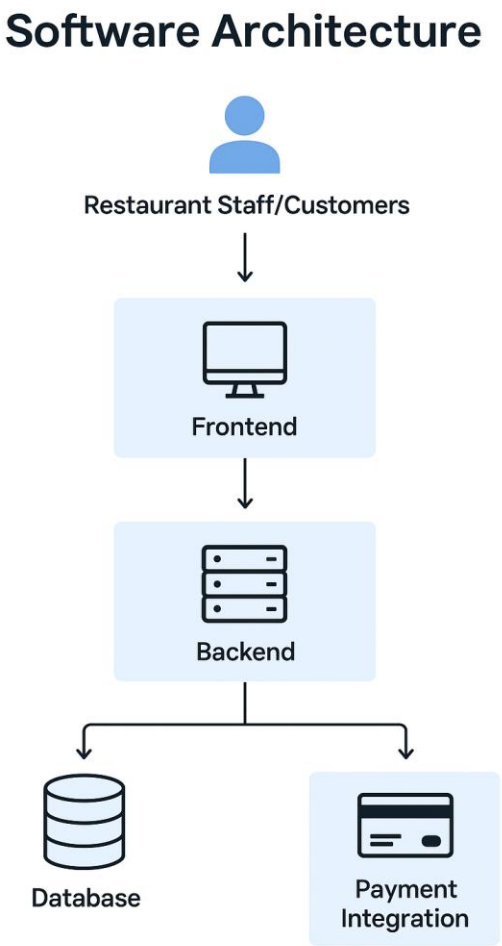
1.3 Definitions and Acronyms

- **POS** – Point of Sale
- **SDS** – Software Design Specification
- **UI** – User Interface
- **API** – Application Programming Interface
- **DBMS** – Database Management System

2. System Overview

The Smart Restaurant POS System provides an end-to-end digital workflow for restaurant operations, from customer ordering to billing, reporting, inventory control, and administrative oversight. The design emphasizes modularity, maintainability, scalability, and secure handling of sensitive data.
The system is implemented using a multi-tier architecture comprising the presentation layer, business logic layer, and data layer. It supports role-based access control, responsive user interfaces, and both synchronous and asynchronous communication patterns.

3. System Architecture



The system follows a three-tier architecture model:

3.1 Presentation Layer

Handles user interaction and includes interfaces for waitstaff, kitchen staff, cashiers, and administrators. It provides intuitive screen flows, validation, and triggers business logic through exposed APIs.

3.2 Business Logic Layer

Implements the core system operations such as order management, billing, tax computation, inventory adjustments, and reporting. This layer encapsulates all processing rules, ensuring consistency and integrity.

3.3 Data Layer

Uses a relational database to store structured data, including menu items, orders, user accounts, table assignments, and inventory records. The schema is normalized to reduce redundancy and maintain consistency.

4. Design Considerations

4.1 Assumptions

- Users have basic computer literacy.
- The restaurant environment includes reliable local networking.
- System users will require authentication before accessing any critical functions.

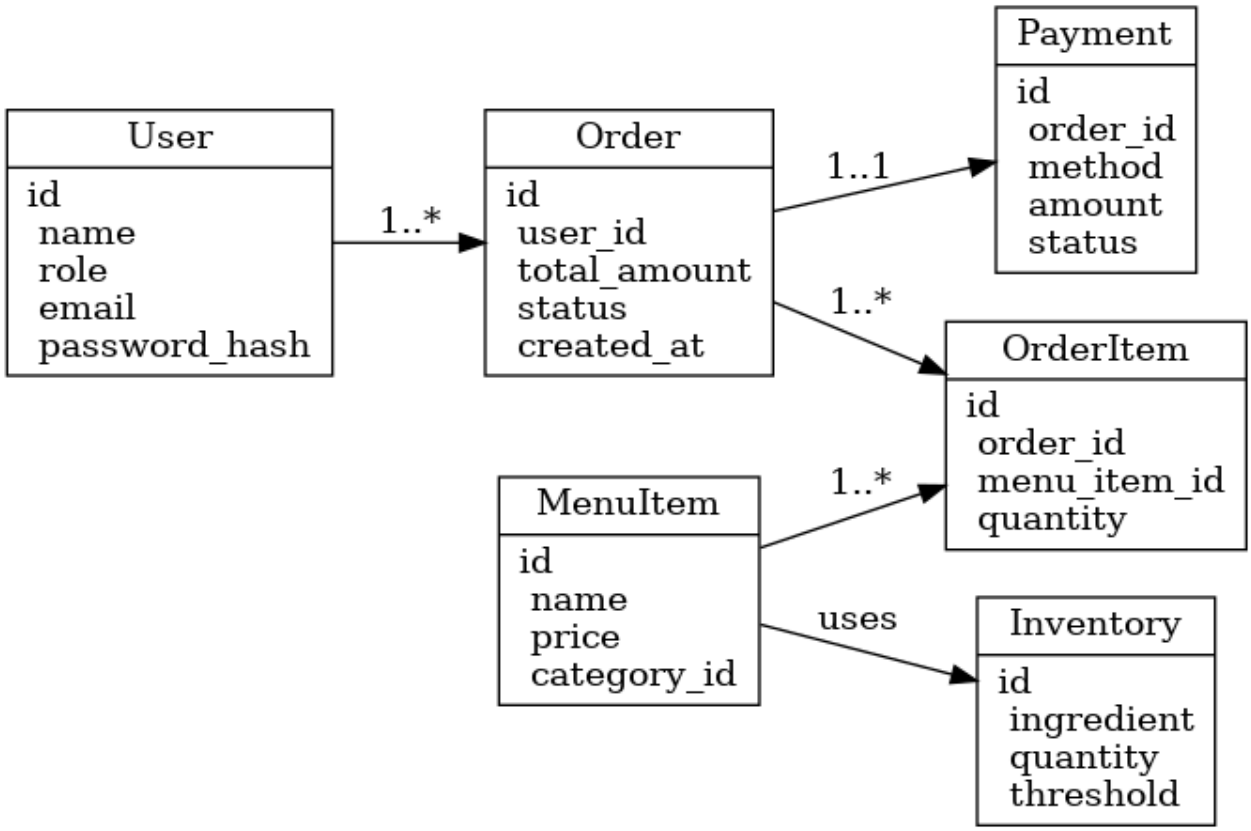
4.2 Constraints

- Must comply with standard data protection requirements.
- Must be operational on typical restaurant hardware (PC terminals, tablets).
- System performance must remain stable during peak hours.

4.3 Goals

- Provide fast and intuitive order processing.
- Maintain accurate, real-time inventory levels.
- Enable secure login and role-based permissions.
- Support robust reporting and analytics.

4.4 Entity Relationship Diagram (ERD)



5. System Modules

5.1 User Management Module

Responsible for user registration, authentication, authorization, password management, and activity logging. Implements role-based permissions such as Admin, Manager, Waiter, and Cashier.

5.2 Menu Management Module

Allows authorized users to create, update, categorize, and deactivate menu items. Includes pricing, availability toggling, and optional modifiers.

5.3 Order Processing Module

Handles order creation, modification, cancellation, and routing to the kitchen. Supports table-based and takeaway ordering, split billing, and special instructions.

5.4 Billing and Payment Module

Generates receipts, handles different payment modes (cash, card, mobile), applies discounts, and records daily financial summaries.

5.5 Inventory Management Module

Tracks ingredient usage, stock levels, reorder thresholds, supplier details, and automated deductions based on fulfilled orders.

5.6 Reporting and Analytics Module

Provides daily, weekly, and monthly performance reports, including sales trends, inventory movements, and employee activity summaries.

6. Data Design

6.1 Database Strategy

A relational database is selected to ensure transactional integrity and support complex queries. Primary and foreign keys maintain relationships between entities.

6.2 Key Entities

- **User**: Stores credentials, roles, and status.
- **Menu Item**: Records item details, category, and price.
- **Order**: Captures the overall order details including timestamps.
- **Order Item**: Links each menu item to its corresponding order.
- **Table**: Represents seating units in the restaurant.
- **Payment**: Stores transactional details of completed orders.
- **Inventory Item**: Tracks ingredient details and stock levels.

7. Component Design

7.1 Interfaces

- **Login Interface**: Authenticate users.
- **Dashboard**: Presents role-specific options.
- **Order Interface**: Used by waitstaff to input customer orders.
- **Kitchen Display Interface**: Shows pending meal preparations.
- **Admin Panel**: Central hub for advanced configuration and reporting.

7.2 Component Interaction

Interaction between components is achieved through well-defined APIs in the business logic layer. All persistent operations route through the database abstraction layer.

8. Workflow Designs

8.1 Order Workflow

1. Waitstaff selects a table.
2. Items are added to the order.
3. The order is submitted to the kitchen.
4. Kitchen staff updates the status to *In Progress* and later *Completed*.
5. Cashier finalizes billing when the customer requests payment.

8.2 Inventory Workflow

1. Admin inputs starting inventory.
2. Each processed order deducts ingredient quantities.
3. When levels reach reorder thresholds, alerts are generated.
4. Admin updates stock upon receiving new supplies.

8.3 User Management Workflow

1. Admin creates user accounts.
2. Users authenticate through the login interface.
3. Permissions determine which modules each user may access.

9. Security Considerations

- All passwords are stored using secure hashing techniques.
- User sessions include timeout and re-authentication requirements.
- Sensitive operations require elevated roles.
- Data access is controlled using least-privilege principles.

10. Testing Strategy

Testing will be performed at multiple levels:

- **Unit Testing**: Validates individual components.
- **Integration Testing**: Ensures inter-module consistency.
- **System Testing**: Confirms the entire system functions as expected.
- **User Acceptance Testing (UAT)**: Final validation by restaurant staff.

11. Conclusion

This SDS outlines the complete design and architectural foundation for the Smart Restaurant POS System. It provides clarity across all components and ensures the system can be reliably developed, maintained, and scaled for future enhancements.